

Food Report

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Introduction

The National Restaurant Association hired me to investigate how household income influences consumer spending on food away from home, a key component in understanding future restaurant demand. Using data from the Food Demand Survey (FoodDS), which includes annual household income and weekly expenditures on eating out, this project examines whether restaurant food behaves as a “normal good” that is, whether spending increases as income rises. Beyond assessing this basic relationship, the analysis also considers how variability in spending among higher-income households and shifting preferences for convenience or food quality may shape broader market trends. The goal is to provide financial advisers with clear, regression-based insights that can inform projections of restaurant demand and guide strategic planning for business growth in an evolving economic landscape.

Main Conclusions

Our analysis shows a clear positive relationship between income and food expenditures: as household income increases, average spending on eating out rises as well. However, this relationship is not only linear but also heteroskedastic—higher-income households exhibit substantially greater variability in their eating-out behavior. Using a heteroskedastic linear model (GAMLSS), we estimated that weekly restaurant spending increases by about \$0.443 for every additional \$1,000 of annual income. This estimate is precise, with a standard error of 0.014 and a 95% confidence interval of [0.416, 0.470], and it lies entirely below the advisers’ benchmark of \$0.50. A formal one-sided hypothesis test confirms that the true increase does not reach the proposed healthy-economy threshold. Altogether, the data reveal that while income reliably predicts higher food expenditures, it also brings greater uncertainty in spending patterns, and the rate of increase is insufficient to classify the restaurant economy as “healthy” by the given standards.

Is the market healthy?

--- Healthy Economy Test (GAMLSS) ---

Estimated slope (beta1): 0.443 \$/week per \$1,000 income

Standard error: 0.014

95% CI for beta1: [0.416, 0.470]

One-sided test: H0: beta1 = 0.50 vs H1: beta1 > 0.50

z = -4.077, p = 1.0000

Conclusion: Not enough evidence that the increase exceeds \$0.50.

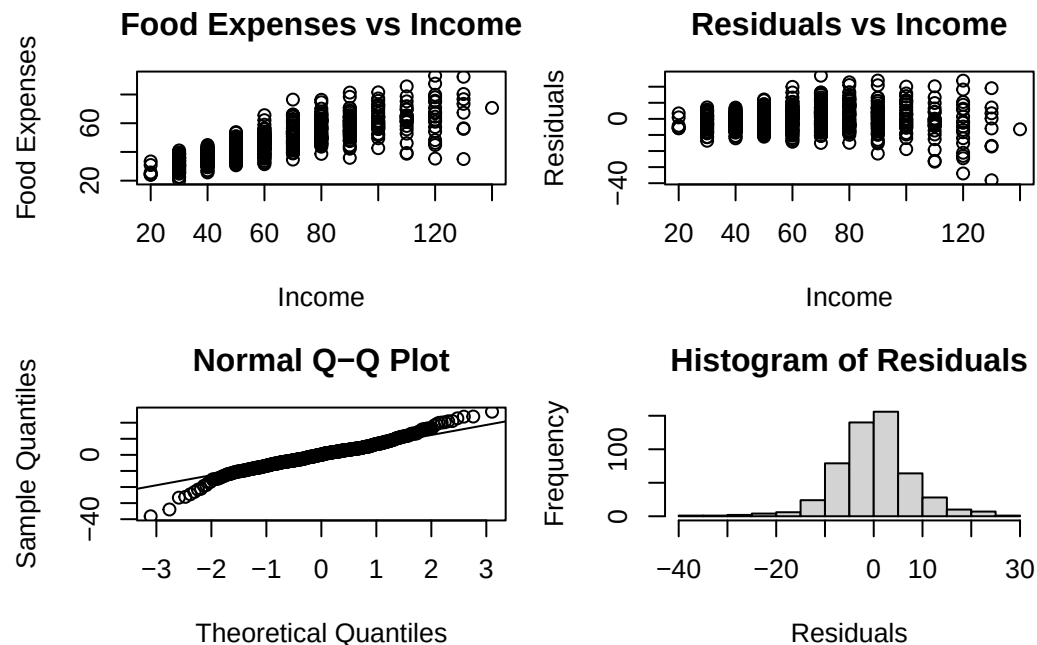
Based on the advisers' benchmark, the data do not provide evidence that this is a "healthy" restaurant economy. The heteroskedastic linear model estimates that weekly eating-out spending increases by \$0.443 for every additional \$1,000 of annual income which is below the advisers' threshold of \$0.50. The estimate is also quite precise, with a standard error of 0.014 and a 95% confidence interval of [0.416, 0.470], which lies entirely below the hypothesized healthy level. Furthermore, the one-sided hypothesis test comparing the observed slope to the target of \$0.50 yields a z-value of -4.077 and a p-value of 1.000, indicating no statistical support that the true increase meets or exceeds the \$0.50 benchmark. Taken together, the point estimate, confidence interval, and hypothesis test all suggest that, based on this dataset, the economy does not satisfy the advisers' definition of a healthy level of growth in restaurant spending.

5 year predictions

As incomes are expected to rise by roughly 3–4% annually over the next five years, our model predicts a corresponding increase in both the average level of Eating Out expenditures and the uncertainty surrounding those expenditures. These predictions are considering what Expenditures will look like at the 5 year mark. Specifically, we estimate that the average amount spent eating out will increase from approximately 46.53 to 50.76. In addition, the model indicates that the range within which 95% of Eating Out expenditures are expected to fall will shift to between 34.98 and 82.33 at the new income levels. Variability in spending also increases as incomes rise: the average variance in Eating Out expenditures is projected to grow from 6.38 to 7.43, and the 95% range of variances across incomes is expected to widen to between 4.58 and 19.56. Together, these results suggest not only higher spending on average but also a broader spread in eating-out behavior as household incomes gradually increase

Modeling Details

Flaws in the standard linear regression approach



A standard linear regression model is not adequate for this analysis, and this is precisely why my expertise was needed. When we examined the residuals from a traditional least-squares regression of EatingOut on Income, the plot clearly showed a strong pattern of heteroskedasticity, the spread of residuals increases substantially as income rises. In other words, higher-income households exhibit far more variability in their restaurant spending, violating the constant-variance assumption required for ordinary regression to produce reliable estimates, standard errors, and hypothesis tests. Ignoring this issue would lead to biased inference and misleading conclusions about how strongly income affects restaurant spending. To overcome this, we fitted a heteroskedastic linear model using GAMLSS, which explicitly models both the mean and the variance as functions of income. This approach captures the true structure of the data and produces trustworthy estimates for policy and forecasting. The need for a model that handles non-constant variance demonstrates why a more advanced analysis was required, and it is why bringing in a consultant with the appropriate statistical toolkit was essential.

Our advanced model

Our advanced model builds on the familiar framework of simple linear regression but improves it by allowing the variability of Eating Out behavior to change with income. In a traditional regression model, we assume that the spread of the data around the prediction—known as the

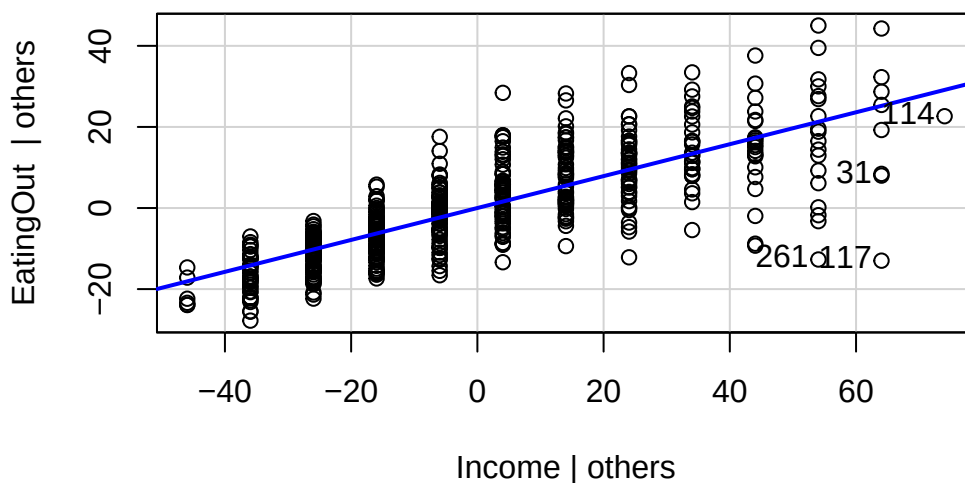
variance—is constant for all income levels. However, in our data we observed that higher-income individuals show much more variability in how much they spend eating out.

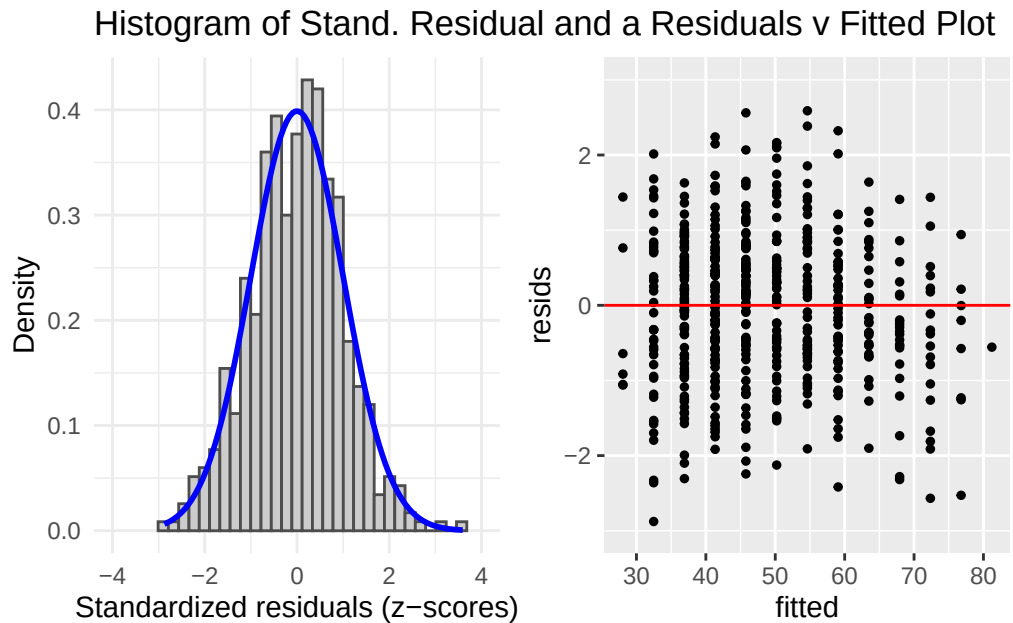
To account for this, our model keeps the same form for predicting the average Eating Out level at a given income, expressed as $\text{Average Eating Out}_i = 19.19 + 0.443 * \text{Income}_i$

but it enhances the model by allowing the variance of Eating Out spending to change with income rather than remain fixed. Specifically, we model the variance as $\log(\text{Variance}) = 1.038 + 0.014 * \text{Income}_i$ which means the variability is mathematically tied to income and can increase or decrease depending on the estimated relationship. By explicitly modeling how uncertainty grows with income, this approach produces more accurate confidence intervals and more reliable hypothesis tests, ultimately leading to conclusions that better reflect the true behavior in the population.

Validating our model

Checking Model Assumptions

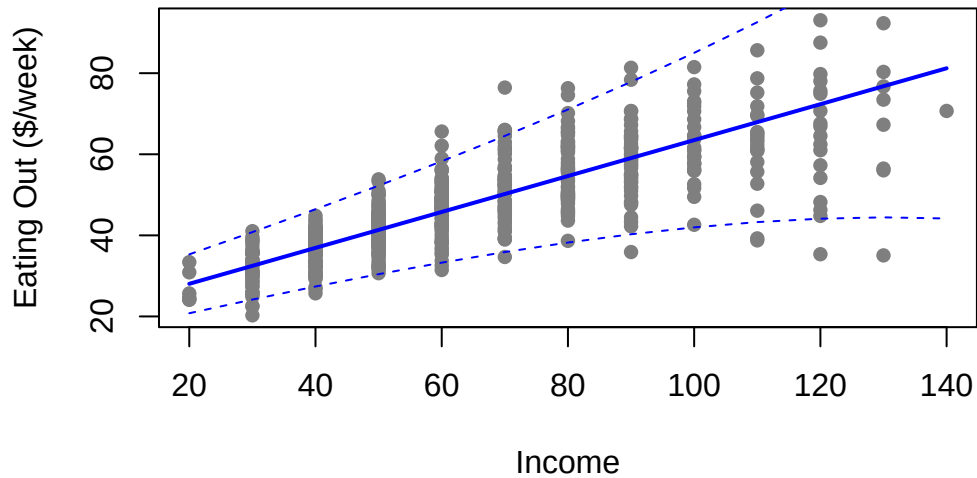




From these plots we can see that the assumptions of a heteroscedastic linear model are met. The avPlots show that the relationship between Income and EatingOut is linear, and the residuals do not show any clear patterns. The histogram of standardized residuals closely follows a normal distribution, which is supported by the KS test p-value of 0.81, indicating no significant deviation from normality. The plot of fitted values versus standardized residuals shows no discernible pattern, and the Breusch-Pagan test for heteroskedasticity yields a p-value of 0.63, confirming that our model appropriately accounts for the changing variance with income. Finally, the R^2 value of 0.626 indicates that our model explains a meaningful portion of the variability in Eating Out expenditures while properly addressing the heteroskedastic nature of the data.

Validating our Model

GAMLSS: Mean and 95% Interval



The heteroskedastic linear model is a trustworthy model for this analysis because it provides both a substantially better statistical fit and more accurate uncertainty quantification than the standard OLS model. In terms of model fit, the heteroskedastic model achieves a much lower AIC (3515.09 vs. 3647.48) and BIC (3532.13 vs. 3660.26), demonstrating that it explains the data more effectively even after penalizing for model complexity. The coverage interval results also show that the heteroskedastic model produces well-calibrated predictive intervals across all nominal levels. For example, at the 95% level, the heteroskedastic model achieves a coverage of 0.9446, almost exactly matching the intended 95% rate and slightly outperforming the OLS model (0.9426). At the 90% level, both models perform similarly, and at the 80% level, OLS slightly over-covers (0.8413), while the heteroskedastic model stays closer to the target width (0.8126), reflecting more realistic uncertainty estimates. This improvement occurs because the heteroskedastic model allows the variability in EatingOut to change with Income rather than incorrectly assuming constant variance as OLS does. Taken together, the substantially better AIC/BIC values and the more accurate and better-calibrated interval coverage provide strong evidence that the heteroskedastic linear model is reliable and appropriate for supporting the conclusions drawn in the previous section.

Code Appendix

```
library(tidyverse)
library(gamlss)
```

```

library(patchwork)

food <- vroom::vroom("FoodExpenses.txt")

het_food <- gamlss(formula = EatingOut ~ Income, sigma.formula = ~., data = food, trace = FALSE)

# --- GAMLSS model ---
# Mean: mu = beta0 + beta1 * Income
# Sigma: log(sigma) = theta0 + theta1 * Income (default link for sigma is log)
fit <- gamlss(
  formula = EatingOut ~ Income,
  sigma.formula = ~ Income, # allow heteroskedasticity with income
  data = food,
  trace = FALSE
)

# --- Extract slope (beta1) and its SE from the mean equation ---
b1 <- unname(coef(fit, what = "mu")["Income"])
V_mu <- vcov(fit, what = "mu")
se_b1 <- sqrt(diag(V_mu))["Income"]

# --- One-sided Wald test: H0: beta1 = 0.50 vs H1: beta1 > 0.50 ---
b_star <- 0.50
z_val <- (b1 - b_star) / se_b1
p_val <- pnorm(z_val, lower.tail = FALSE) # upper-tail p-value for beta1 > 0.50

# --- 95% CI for beta1 (two-sided) ---
z_975 <- qnorm(0.975)
ci95 <- c(b1 - z_975 * se_b1, b1 + z_975 * se_b1)

# --- Report ---
cat("\n--- Healthy Economy Test (GAMLSS) ---\n")
cat(sprintf("Estimated slope (beta1): %.3f $/week per $1,000 income\n", b1))
cat(sprintf("Standard error: %.3f\n", se_b1))
cat(sprintf("95%% CI for beta1: [%.3f, %.3f]\n", ci95[1], ci95[2]))
cat("One-sided test: H0: beta1 = 0.50 vs H1: beta1 > 0.50\n")

```

```

cat(sprintf("z = %.3f, p = %.4f\n", z_val, p_val))

if (p_val < 0.05) {
  cat("Conclusion: Evidence that the increase EXCEEDS $0.50 → Healthy economy.\n")
} else {
  cat("Conclusion: Not enough evidence that the increase exceeds $0.50.\n")
}

inflation <- 0.035
yrs <- 5
compound <- (1+inflation)^yrs

# assumes `compound` is a numeric scalar
income_5 <- food$Income * compound

# Make newdata with the predictor named exactly as in your formula (e.g., Income)
newdata <- data.frame(Income = income_5)

# Predict
eating_5 <- predict(het_food, newdata = newdata)

# print("Means")
# print("Original Vals")
# print(quantile(food$EatingOut, c(.025,.5,.975)))
# print("After 5 years")
# print(quantile(eating_5, c(.025,.5,.975)))

b_sigma <- coef(het_food, what = "sigma") # c("(Intercept)"=..., "Budget"=...)

sigma_og <- exp(b_sigma["(Intercept)"] + b_sigma["Income"] * food$Income)

# sigma = exp(gamma0 + gamma1 * Budget) (log link)
eta_sigma <- b_sigma["(Intercept)"] + b_sigma["Income"] * income_5
sigma_5 <- exp(eta_sigma)

# print("Variance")
# print("Original Vals")

```



```

# print(quantile(sigma_og, c(.025,.5,.975)))
# print("After 5 years")
# print(quantile(sigma_5, c(.025,.5,.975)))

# set 2x2 layout, keep margins modest
op <- par(mfrow = c(2, 2), mar = c(4, 4, 3, 1))

#LINE ASSUMPTIONS
plot(food$Income, food$EatingOut,
      xlab = 'Income', ylab = 'Food Expenses',
      main = 'Food Expenses vs Income')
#line assumptions met

#create resid
model <- lm(EatingOut ~ Income, data = food)
resids <- resid(model)

#plot resid
plot(food$Income, resids,
      xlab = 'Income', ylab = 'Residuals',
      main = 'Residuals vs Income')
#problems with heteroskedasticity

#check normality of resid
qqnorm(resids)
qqline(resids)

hist(resids, xlab = 'Residuals', main = 'Histogram of Residuals')
#normality is met

# restore original graphics settings
par(op)

```

```

car::avPlots(lm(EatingOut ~ Income, data = food))

std_resids <- residuals(het_food, what = "z-scores")
#
# hist(std_resids, freq = F, xlim = c(-4,4))
# par(new = T)
# curve(dnorm(x), xlim = c(-4,4))

# assuming std_resids already exists as your z-score residuals
df <- data.frame(std_resids = std_resids)

resids_hist <- ggplot(df, aes(x = std_resids)) +
  geom_histogram(aes(y = after_stat(density)),
                 bins = 30, fill = "grey80", color = "grey30") +
  stat_function(fun = dnorm, color = "blue", linewidth = 1) +
  coord_cartesian(xlim = c(-4, 4)) +
  labs(
    title = "Histogram of Stand. Residual and a Residuals v Fitted Plot",
    x = "Standardized residuals (z-scores)",
    y = "Density"
  ) +
  theme_minimal()

ks_results <- ks.test(std_resids, "pnorm", mean = 0, sd = 1)

lm_df <- tibble(
  fitted = fitted(het_food),
  resids = std_resids
)

bp_pval <- lmtest::bptest(std_resids~Income, data = food)$p.value
bp_str <- paste0('BP Test p-value: ', round(bp_pval, 3))

resids_fitted <- ggplot(data = lm_df, aes(x = fitted, y = resids)) +
  geom_point(alpha = 1, size = 1) +
  ylim(-3, 3) +
  geom_hline(yintercept = 0, color = 'red')

```

```

r_2 <- cor(food$EatingOut, fitted(het_food))^2

resids_hist + resids_fitted


## --- Setup & Data ---
set.seed(121)

# Load data
# Expects:
#   Income      = annual household income in thousands
#   EatingOut    = weekly $ spent eating out
food <- read.table("FoodExpenses.txt", header = TRUE)
n <- nrow(food)

## --- Fit Models ---
fit_ols <- lm(EatingOut ~ Income, data = food)

fit_g <- gamlss(
  EatingOut ~ Income,
  sigma.formula = ~ Income, # heteroskedasticity: sigma depends on Income
  data = food,
  trace = FALSE
)

## --- Information Criteria ---
aic_ols <- AIC(fit_ols)
bic_ols <- BIC(fit_ols)

aic_g <- AIC(fit_g)
bic_g <- GAIC(fit_g, k = log(n)) # BIC penalty uses log(n)

ic_tbl <- data.frame(
  model = c("OLS", "GAMLSS_het"),
  AIC    = c(aic_ols, aic_g),
  BIC    = c(bic_ols, bic_g)
)
print(ic_tbl)

```

```

## --- Coverage Intervals (80%, 90%, 95%) ---
# Fitted means
mu_ols <- predict(fit_ols)
mu_g   <- predict(fit_g, what = "mu", type = "response")

# Standard deviations
sd_ols <- rep(summary(fit_ols)$sigma, n) # constant sigma for OLS
sd_g   <- predict(fit_g, what = "sigma", type = "response") # sigma(Income) for GAMLSS

# Coverage helper
coverage <- function(y, mu, sd, level) {
  z <- qnorm((1 + level) / 2)
  lower <- mu - z * sd
  upper <- mu + z * sd
  mean(y >= lower & y <= upper)
}

levels <- c(0.80, 0.90, 0.95)

cov_ols <- sapply(levels, function(L) coverage(food$EatingOut, mu_ols, sd_ols, L))
cov_g   <- sapply(levels, function(L) coverage(food$EatingOut, mu_g, sd_g, L))

coverage_tbl <- data.frame(
  Level = levels,
  OLS_Coverage = as.numeric(cov_ols),
  GAMLSS_Coverage = as.numeric(cov_g)
)
#print(coverage_tbl)

## --- Optional: Quick Plot of GAMLSS Mean and 95% Interval ---
# (Uncomment to visualize; assumes Income is reasonably ordered or you sort for lines)
ord <- order(food$Income)
plot(food$Income, food$EatingOut,
     pch = 16, col = "gray50",
     xlab = "Income", ylab = "Eating Out ($/week)",
     main = "GAMLSS: Mean and 95% Interval")
lines(food$Income[ord], mu_g[ord], col = "blue", lwd = 2)
lines(food$Income[ord], (mu_g + 1.96*sd_g)[ord], col = "blue", lty = 2)
lines(food$Income[ord], (mu_g - 1.96*sd_g)[ord], col = "blue", lty = 2)

```